OLIVIA SAMOLEJ

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EDUCATION

University of Illinois at Urbana-Champaign Expected May 2027
Bachelor of Science in Atmospheric Science GPA: 3.82/4.00

SKILLS

Python, Java, ASL (Proficient)

EXPERIENCE

University of Illinois Urbana-Champaign Champaign, IL
Undergraduate Researcher January 2024 - Present
Research Advisor: Dr. Robert Rauber

- Quantify and understand the microphysical processes that occur at the termination of the radar bright band that is associated with melting hydrometeors within winter cyclones using data collected during the NASA IMPACTS (Investigation of Microphysics and Precipitation in the Atlantic Coast Threatening Snowstorms) field campaigns
- Analyze the rain/sleet/snow transition, determining if mesoscale circulations develop within the transition region and how those circulations impact the distribution of those 3 species of hydrometeors
- Simulate the melting-layer bright band using the Weather Research and Forecasting (WRF) model and then compare it with data from the ER2 X-band Doppler Radar, the High-Altitude Imaging nadir pointing Wind and Rain Airborne Profiler, and the nadir pointing W-band Cloud Radar System

University of Illinois Urbana-Champaign Champaign, IL
Teaching Assistant ATMS 120 – Severe and Hazardous Weather August 2025 - Present

- Assisted with teaching and grading in a course with over 500 students
- Explained and communicated complex mathematical concepts to the general campus population
- Held frequent office hours to provide additional support and improve student understanding

University of Michigan Ann Arbor, MI
NSF PICASSO REU Researcher May 2025 – July 2025
Research Advisor: Dr. Jeremy Basis

- Conducted research on Antarctic ice shelves to evaluate their stability and implications for future sea level rise
- Utilized the BISICLES ice sheet model to simulate damage to the Thwaites, Pine Island, and Dotson/Crosson ice shelves, comparing a linear and non-linear damage model
- Determined that the non-linear damage model more accurately captured observed ice shelf behavior

Geophysical Fluid Dynamics Laboratory **Princeton**, NJ
Hollings Scholar Researcher January 2026 – Present
Research Advisor: Dr. Veeshan Narinesingh

- Studying quasi-stationary waves and their influence on weather extremes such as heatwaves and cold snaps
- Help identify model biases to improve the representation of large-scale atmospheric circulation in climate models

Professional Membership

ISTORM (Illinois Student Organization for Meteorology) Champaign, IL
Treasure August 2023 - Present

- Organized creative fundraising initiatives, raising over $900 in support of relief efforts for Hurricanes Milton and Helene
- Founded and led a fundraising committee, increasing member engagement and expanding organizational resources
- Represented ISTORM at UIUC’s Engineering Open House, educating grade-school students about atmospheric science; awarded Best Undergraduate Research and 3rd Place Most Engaging

CliMAS Ambassador Champaign, IL

Undergraduate Student Ambassador August 2025 - Present

- Guide prospective students by answering questions and offering honest insights about atmospheric science
- Lead tours of the Natural History Building and assist with recruitment events such as LAS LiftOff and the Majors and Minors Fair

Awards

Ernest F. Hollings Scholarship Recipient

- Ernest F. Hollings Undergraduate Scholarship recognizes outstanding students studying in NOAA mission Fields April 2025

Department of Climate, Meteorology & Atmospheric Sciences Faculty Scholarship (Merit based) Champaign, IL

- Awarded to a rising sophomore in the Atmospheric Sciences Bachelor's degree program who has demonstrated exceptional academic performance as a freshman April 2024

Dean's List

- Fall 2023, Spring 2024, Fall 2025

Presentations

Samolej, O. Bassis, J. Modeling damage evolution across Antarctic ice shelves, School of Earth Society, and Environment Research Review, Urbana-Champaign, IL, February 2026.

Samolej, O. Zaremba, T. Rauber, R. Radar Bright Bands and Finescale Motions Near the Rain–Snow Boundary: Integrating IMPACTS Observations with WRF Simulations, CliMAS Seminar, Urbana-Champaign, IL, February 2026.

Samolej, O. Bassis, J. Modeling damage evolution across Antarctic ice shelves, 106th AMS Annual Meeting, Houston, TX, January 2026.

Samolej, O. Zaremba, T. Rauber, R. Characteristics of the Radar Bright band in Winter Storms, updated, 41st International Conference on Radar Meteorology, Toronto, CA, August 2025.

Samolej, O. Bassis, J. Modeling damage evolution across Antarctic ice shelves, PICCASO REU Research Review, Ann Arbor, MI, July 2025.

Samolej, O. Rauber, R. Zaremba, T. Characteristics of the Radar Bright band in Winter Storms, School of Earth Society, and Environment Research Review, Urbana-Champaign, IL, February 2025.